



BSc (Hons) COMPUTER SCIENCE · FINAL YEAR RESEARCH PROJECT

Explainable Deep Learning Framework for Pneumonia Detection and Classification Using Transfer Learning and Grad-CAM

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Two deficiencies persist across the reported systems

General problem · trust tracks confidence, not reliability

- Automated support is trusted in proportion to how confident it appears rather than how reliable it is.
- Networks are confidently wrong, and calibration does not fix it.
- The consequence is a system that transfers risk to the patient while appearing to reduce it.

Specific problem · granularity and the closed world

- Most published chest radiograph systems are binary, so they cannot support the antibiotic decision.
- Softmax is closed world: a photograph, or a radiograph of an untrained condition, still returns a confident diagnosis.
- Post hoc detection methods are mature, yet none has been carried into an evaluated chest radiograph system.

The property this project is built around

Softmax redistributes probability across the classes it was trained on, for every input it is ever shown. It expresses relative preference among those classes and never the absence of evidence for all of them.

Research question

To what extent can a post hoc and explainable framework that is aware of out of distribution inputs improve the clinical safety of automated four class pneumonia classification from chest radiographs, without retraining the underlying classifier?

RQ1

Performance and error structure

What classification performance, and what per class error structure, do DenseNet121, EfficientNetV2B3 and EfficientNetV2S achieve under a common dataset and split?

RQ2

Input validity

Can inference time signals from an already trained classifier detect out of distribution inputs reliably enough to justify withholding a diagnosis, and at what cost to genuine scans?

RQ3

Explanation localisation

Do the Grad-CAM explanations fall on the lung fields and on the regions clinically associated with each class?

Aim and objectives

Aim

To design, implement and evaluate an explainable four class framework that is aware of out of distribution inputs, to deliver it as a working clinical decision support platform, and to produce evidence about its performance, its input validity behaviour and where its explanations place their attention.

RO1 **Identify** the limitations of existing systems through a critical review, and derive design requirements from them

RO2 **Analyse** three architectures under a common dataset, split and transfer learning protocol, with per class metrics

RO3 **Design and develop** a post hoc safeguard that assesses input validity without retraining, integrated end to end

RO4 **Evaluate** the safeguard and the explanation: retention, separability, and where attention falls

RO5 **Recommend** the conditions under which such a framework is, and is not, suitable for screening and triage

No reviewed system checks input validity

Study	Classes	Explainability	Input validity check	Deployed
Rajpurkar et al. [9]	Pneumonia vs rest	CAM, qualitative	None reported	No
Apostolopoulos & Mpesiana [12]	Binary and three class	None reported	None reported	No
Ozturk et al. [13]	Binary and three class	Heatmaps, qualitative	None reported	No
Narin et al. [14]	Binary COVID-19	None reported	None reported	No
Chowdhury et al. [7]	Normal, viral, COVID-19	Score-CAM, qualitative	None reported	No
Huang & Liao [23]	Multi class chest disease	Limited	None reported	No
This study	Four class, bacterial vs viral	Grad-CAM from logits, lung restricted, quantified	Mahalanobis, energy and MSP, evaluated	Yes, full platform

Three derived requirements separate this design from every row above

DR4 the system must be able to refuse to answer

DR5 several validity signals compared, not one adopted

DR8 explanation localisation measured and disclosed

Research design

PARADIGM

Post positivist

Accuracy, recall, AUROC and in lung attention are objective quantities that can be estimated but not known exactly. Every measurement is reported with its sample size, and a disconfirming result is reported in the same terms as a confirming one.

APPROACH

Predominantly deductive

The closed world property of softmax and the feature space method of Lee et al. yield a testable expectation, which was operationalised into thresholds and measured. The inductive element is kept separate.

STRATEGY

Design science

The question asks to what extent a safeguard can improve safety, and that quantity does not exist until the safeguard has been built. Evaluation within it is experimental.

The freeze point

Calibration follows model selection and uses only training and validation data. Evaluation follows the freeze, so no result reported later was used to adjust the artefact that produced it.

One stratified split, reused by every model and by calibration

9,209

RADIOGRAPHS AFTER AUDIT

duplicate, corrupted and low resolution files removed first

70 / 15 / 15

STRATIFIED SPLIT

6,446 train · 1,381 val · 1,382 test

4

CLASSES

Normal, COVID-19, bacterial, viral

Class	Total	Train	Validation	Test
COVID-19	1,281	897	192	192
Normal	3,271	2,289	491	491
Bacterial pneumonia	3,001	2,101	450	450
Viral pneumonia	1,656	1,159	248	249
Total	9,209	6,446	1,381	1,382

Design consequences

- The audit precedes the split, because a duplicate spanning a partition boundary is a direct route to leakage.
- The realised split file is reused by all three architectures and by calibration, so results are comparable.
- The imbalance is addressed with stratified sampling and a class weighted loss.

Secondary data: curated public research collections, released for research use and carrying no patient identifiers. No human participants, so institutional ethics approval was not applicable.

The safeguard: fitted from genuine data, no retraining

DEPLOYED POLICY

Mahalanobis

feature space

Class conditional Gaussians on the 256-d penultimate features with a single tied covariance, shrunk by 0.01 on the identity. Score is the distance to the nearest class mean.

REPORTED

Energy

output space

The negative log sum exp of the logits at temperature 1.0. Computed and reported for every scan.

REPORTED

Maximum softmax

output space

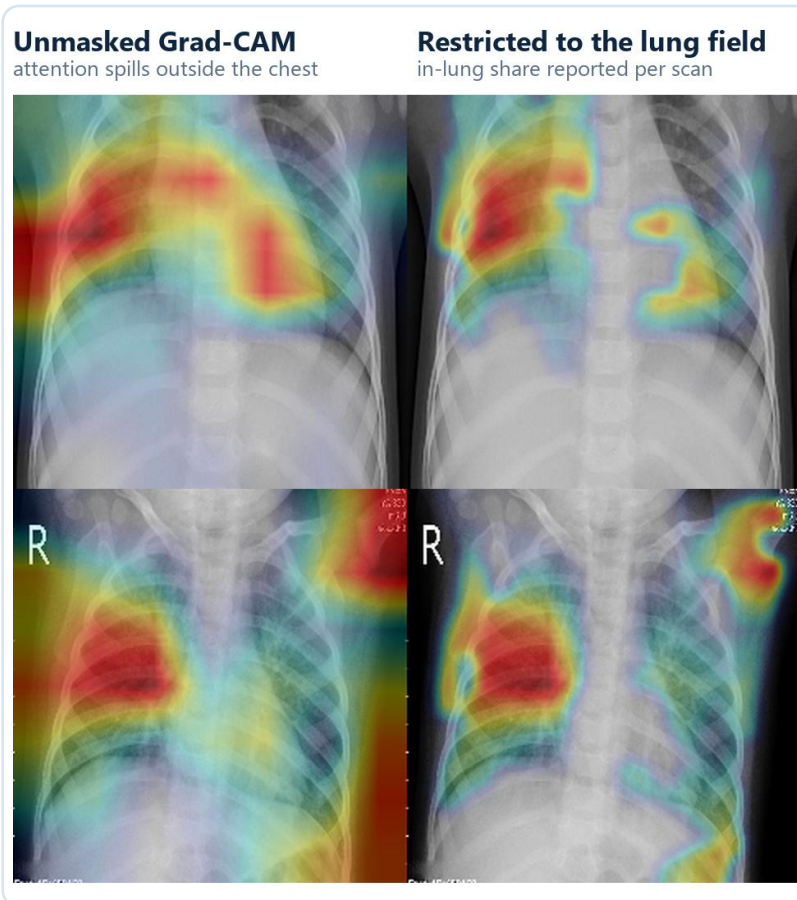
The Hendrycks and Gimpel baseline. Computed and thresholded, ordering reversed because higher means more familiar.

Calibration runs once per model, on genuine data only

- Statistics are fitted on 1,200 training images and thresholded on 600 validation images, at a target retention of 95 per cent of genuine scans.
- The covariance is tied across classes because the fitting sample is only about 4.7 times the 256 dimensions, so four separate matrices would be rank deficient.
- The safeguard never sees an invalid example during calibration. The operating point is set entirely from real radiographs.

When the verdict fires the response carries no diagnosis: no predicted class, no confidence, no probability bars.

What lung restriction changes



Verification runs retained in the inference service workspace.

Read left to right

- Left: the raw map, with substantial mass on the image border and outside the chest entirely.
- Right: the same map restricted to the estimated lung field.
- The measurement is taken before masking, so the restriction cannot flatter the number.

The reason the number is disclosed

A plausible looking heatmap is not evidence that lung tissue was used. Masking made the maps look more convincing without changing the classifier at all, which is exactly why the measurement was added alongside it.

Three architectures, one split, one protocol

Model	Accuracy	Macro precision	Macro recall	Macro F1	Weighted F1	Errors
DenseNet121 · deployed	85.31 %	0.843	0.845	0.844	0.851	203
EfficientNetV2B3	84.30 %	0.834	0.825	0.828	0.838	217
EfficientNetV2S	80.54 %	0.798	0.812	0.798	0.803	269

The margin is not a finding

- The 1.01 point lead over EfficientNetV2B3 is fourteen images out of 1,382, from a single run.
- The methodology committed in advance to treating a difference at that resolution as inconclusive.

And one row is not an architectural result

- EfficientNetV2S was trained with rescaling to the unit interval, while its ImageNet weights expect the interval from minus one to one.
- Its frozen phase reached 68.07 per cent against 83.20, a gap capacity does not explain. This is disclosed, not buried.

NFR1 asked for at least 85 per cent and is met by 0.31 of a percentage point, a margin narrow enough that a different seed could plausibly have failed it.

Both operating criteria met, and one signal fails

92.5 %

GENUINE SCANS RETAINED

test partition. NFR2 floor is 90 %

100 %

FAR PROBE FLAGGED

all 60 generated images. NFR3 floor is 95 %

1.000

MAHALANOBIS AUROC

against energy at 0.539

Quantity	Value
Feature dimensionality	256
Calibration sample	1,200 training and 600 validation
Retention on validation, where thresholds were set	95.0 %
Retention on the test partition	92.5 %
Far probe flagged	100 % of 60 images
AUROC, Mahalanobis, far probe	1.000
AUROC, energy, far probe	0.539
Deployed decision policy	Mahalanobis

The negative result is the informative one

- Energy is a function of the logits. If a model produces large logits for inputs it has never seen, energy cannot separate them.
- That is exactly the documented overconfidence. Mahalanobis operates before that collapse, in feature space.
- Which signal works has to be measured, not

Stated limitation

Sixty generated images of noise, gradients and solid colour from a fixed seed. A far regime sanity check, not validation against real photographs.

Conclusion

This thesis addressed the gap between a chest radiograph classifier that is accurate and one that is safe to deploy.

85.31 %

TEST ACCURACY

macro F1 0.844 on 1,382 held out images

92.5 %

GENUINE SCANS RETAINED

while refusing every probe image

1.000

MAHALANOBIS AUROC

against energy at 0.539

10 / 10

TEST CASES PASSED

against the running stack

What is new here

- A four class explainable framework in which a post hoc safeguard, fitted without retraining, withholds the diagnosis for inputs the model cannot assess, and says why.
- An explanation restricted to the lung fields whose localisation is measured on the unmasked map and disclosed for every scan.
- Evidence rather than a new architecture: a per class characterisation of the four class task, and detection methods carried into a clinical setting where they are actually evaluated and one of them fails.

The practical value lies in the safeguard rather than in the classifier: it applies to a model already deployed.